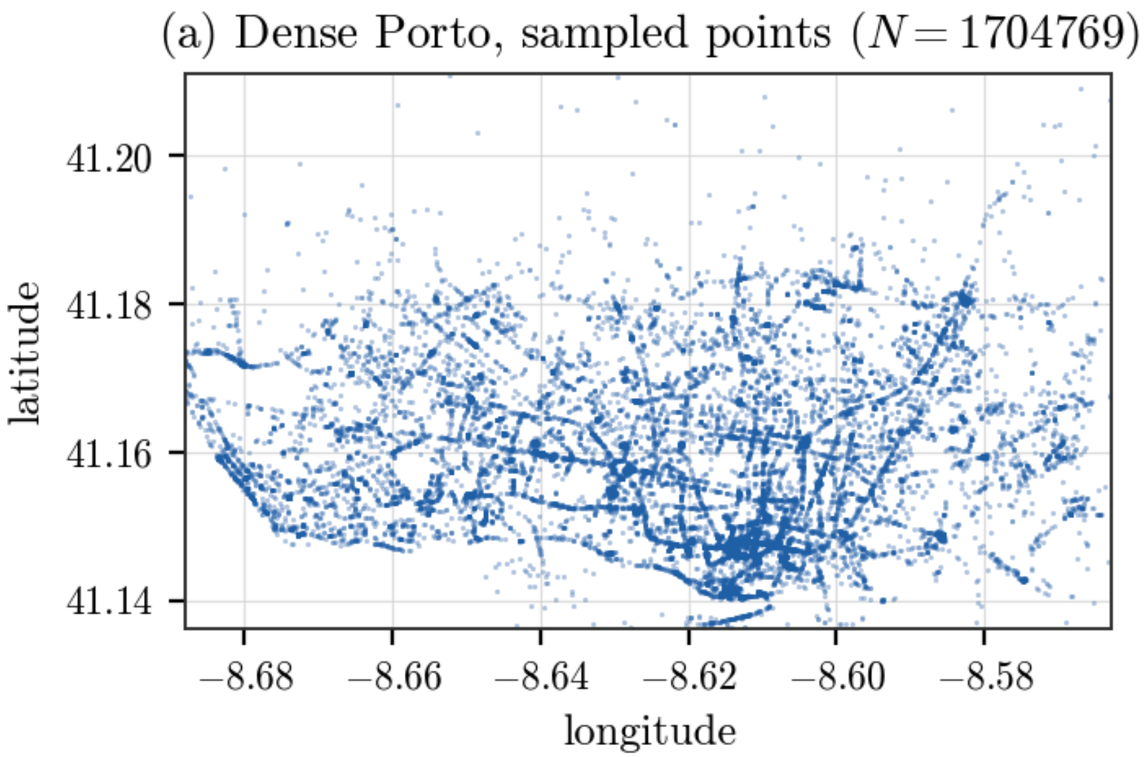
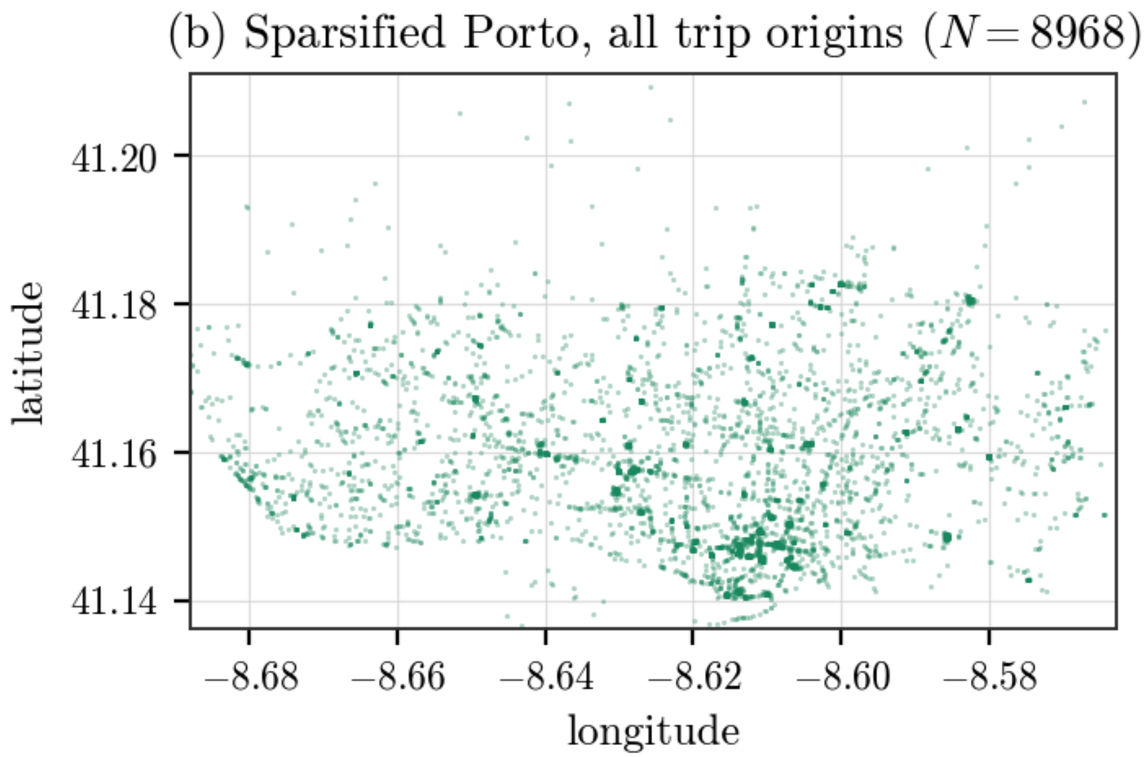
Sparse Porto as a Gurgaon proxy

What matches, what changes, and what SEED actually learns

How Porto's observed trip footprint changes under sparsification :

Question : To what extent does sparsity-calibrated Porto data reproduce the observational characteristics relevant to SEED evaluation on Gurgaon?

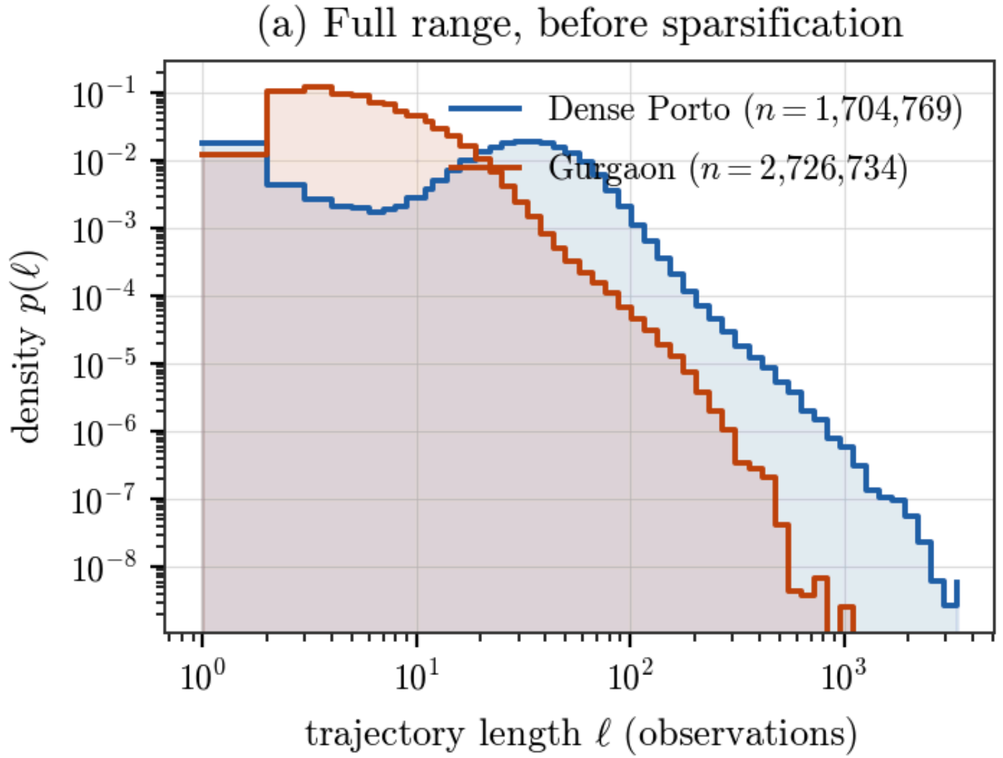


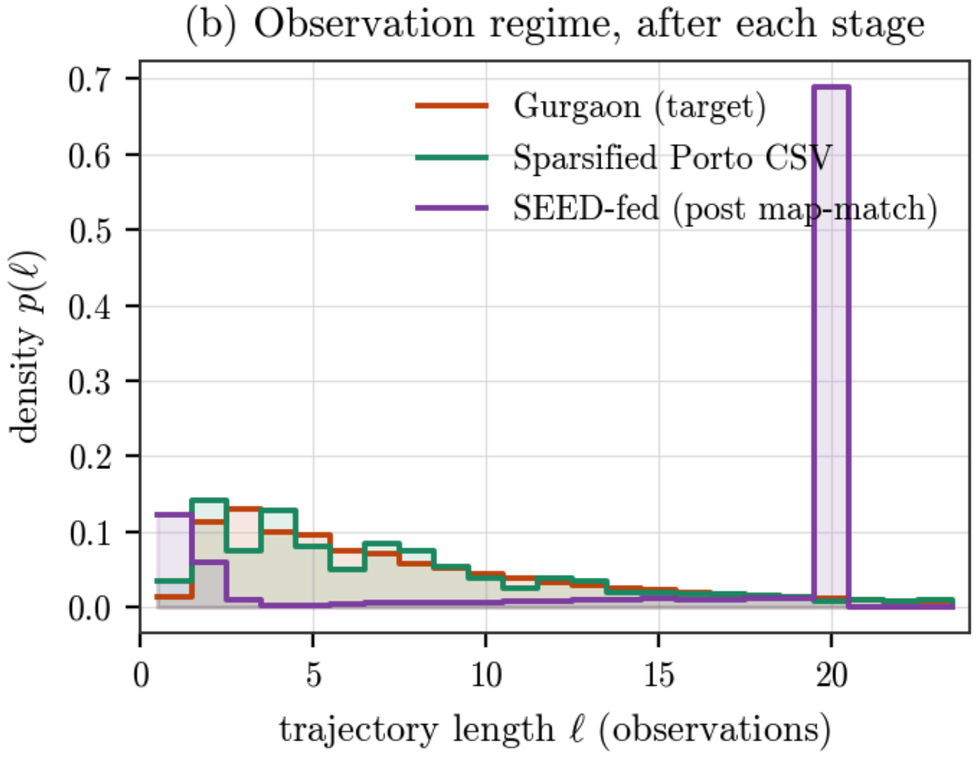


Calibration works at the CSV stage

Trajectory lengths and aggregate sparsity targets align closely before map matching

- **Mean / median trip length**: Gurgaon 9.54 / 7;
  sparse Porto CSV 8.79 / 7.
- **KS distance** to Gurgaon falls from 0.7969 (raw Porto)
  to 0.0497 (sparse CSV).
- **Entity-day density and date coverage land** within about 0.4 percentage points of target;
  singleton rate is higher (3.26% vs 1.21%).

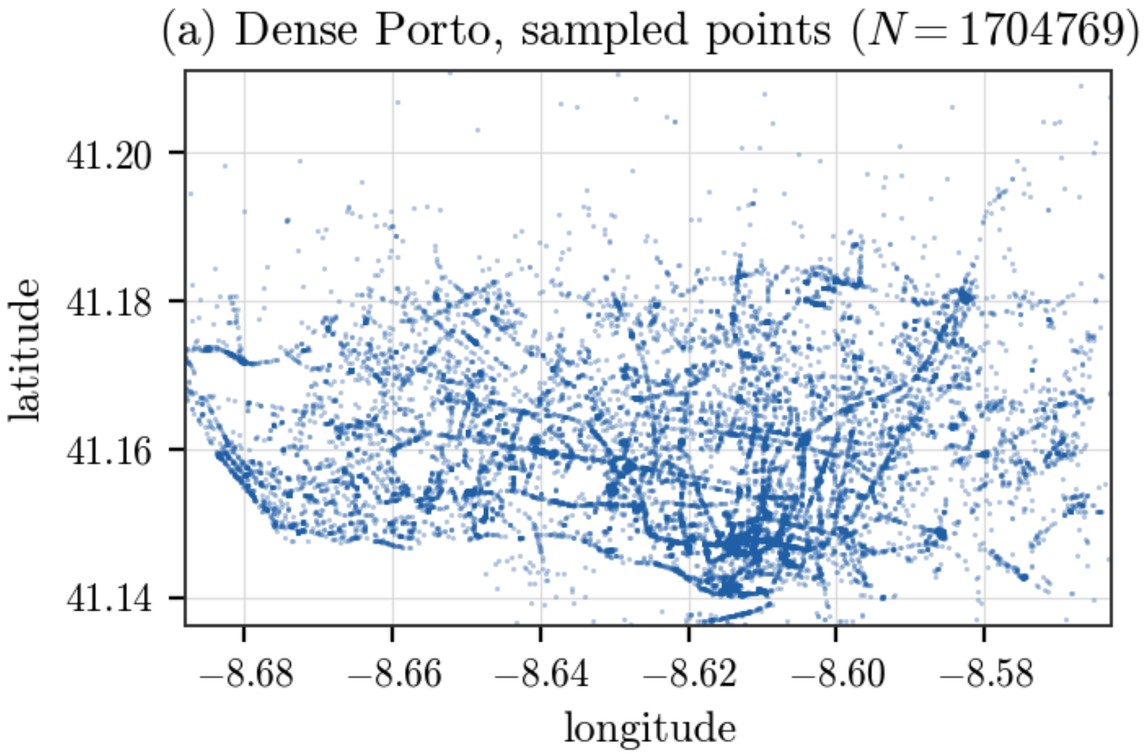


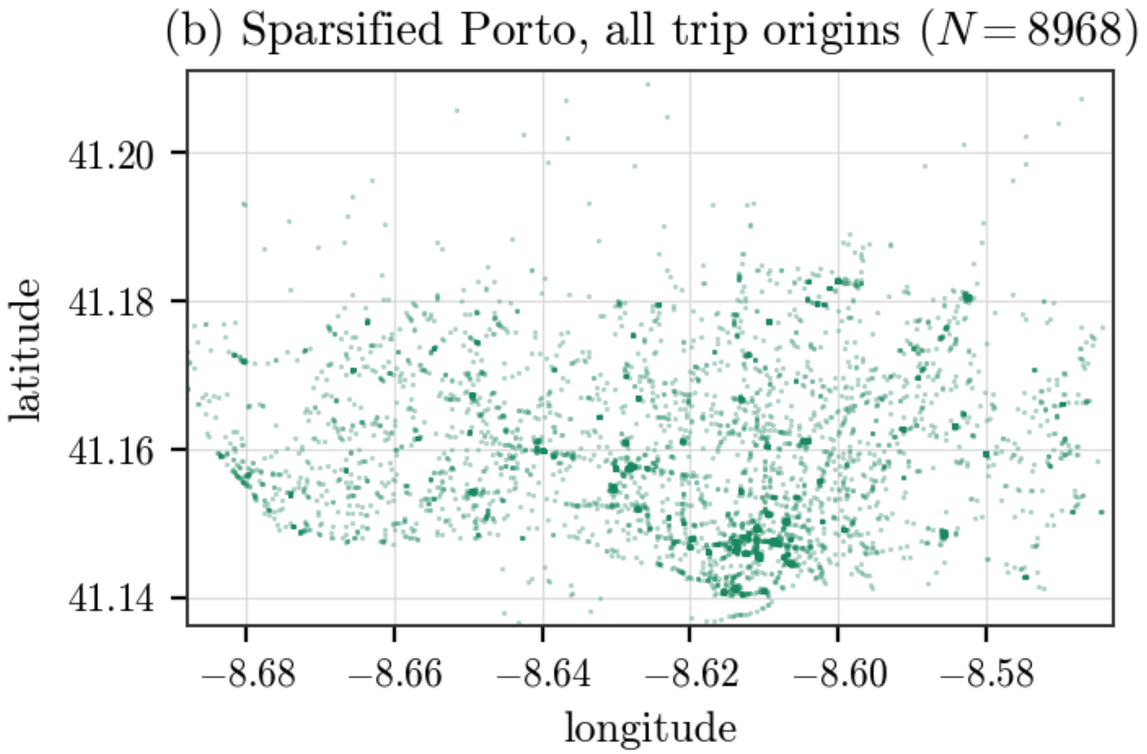


The spatial footprint is thinned, not shifted

Sparsification drops trips while retaining their original coordinates.

- The sparse footprint is a subset of raw Porto's spatial support.
- Thinning is most visible around lower-density periphery areas.
- This checks spatial retention within Porto; it does not compare absolute geographic coverage with Gurgaon.

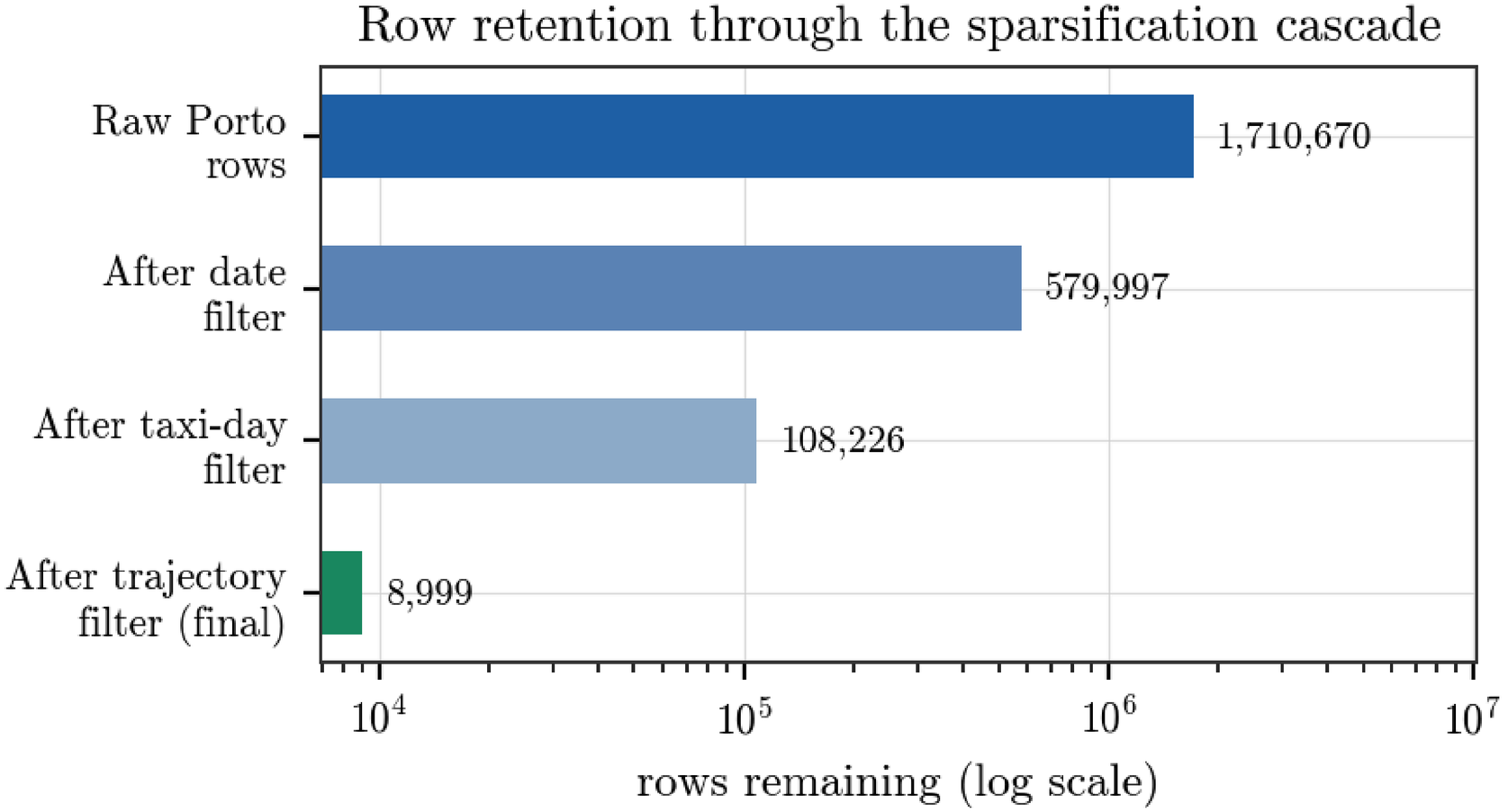




Pipeline retention is dominated by early filters

The date and taxi-day filters remove most trips; the anchor-trip stage reduces the remainder further.

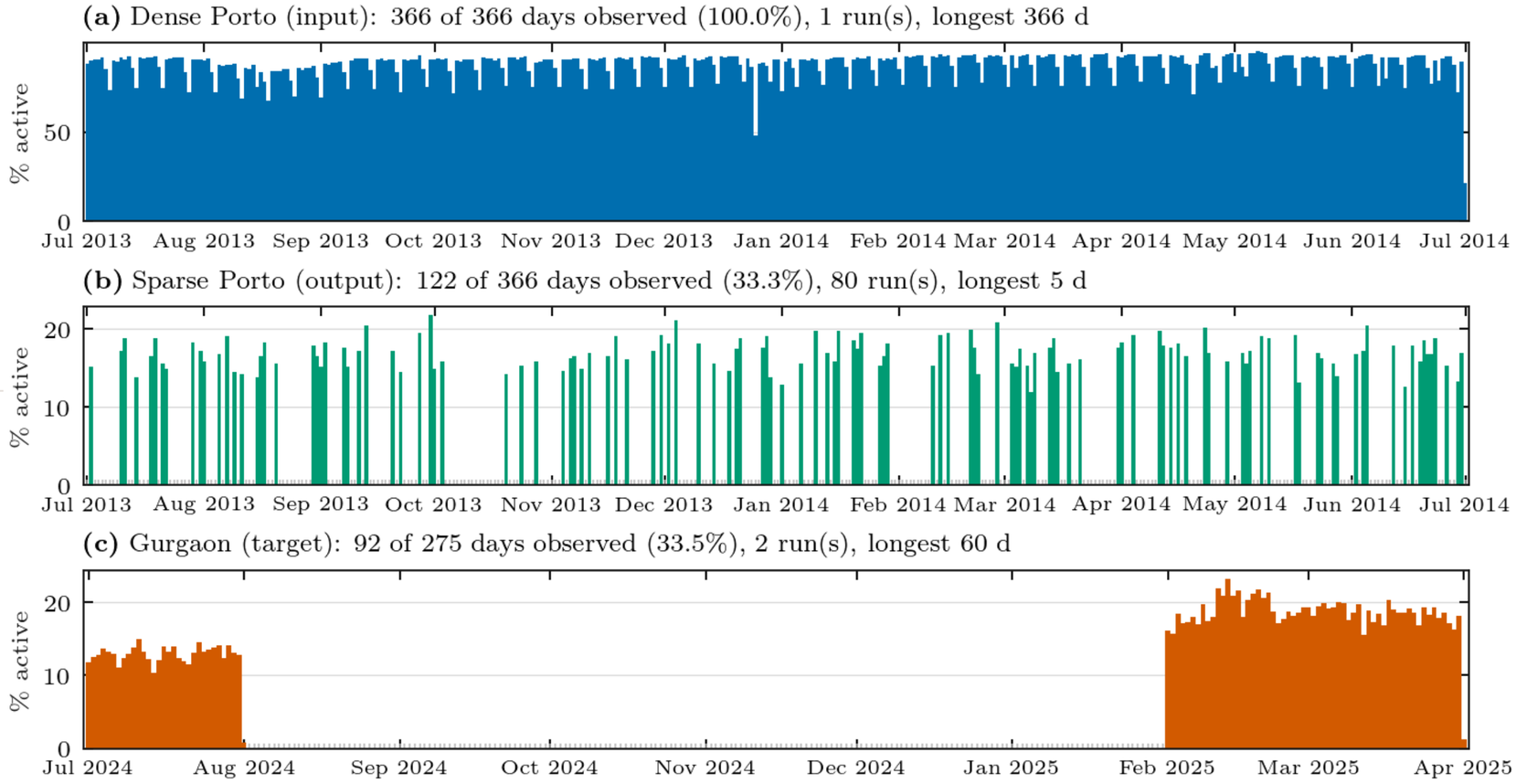
- From 1.71M raw trips, the pipeline keeps about 9,000 output rows.
- Date selection retains 122 of 366 dates; taxi-day selection keeps 8,999 of 48,249 candidate taxi-days.
- The final trajectory filter removes a further 92% of the rows that reach that stage.



Matching totals still leaves structural gaps

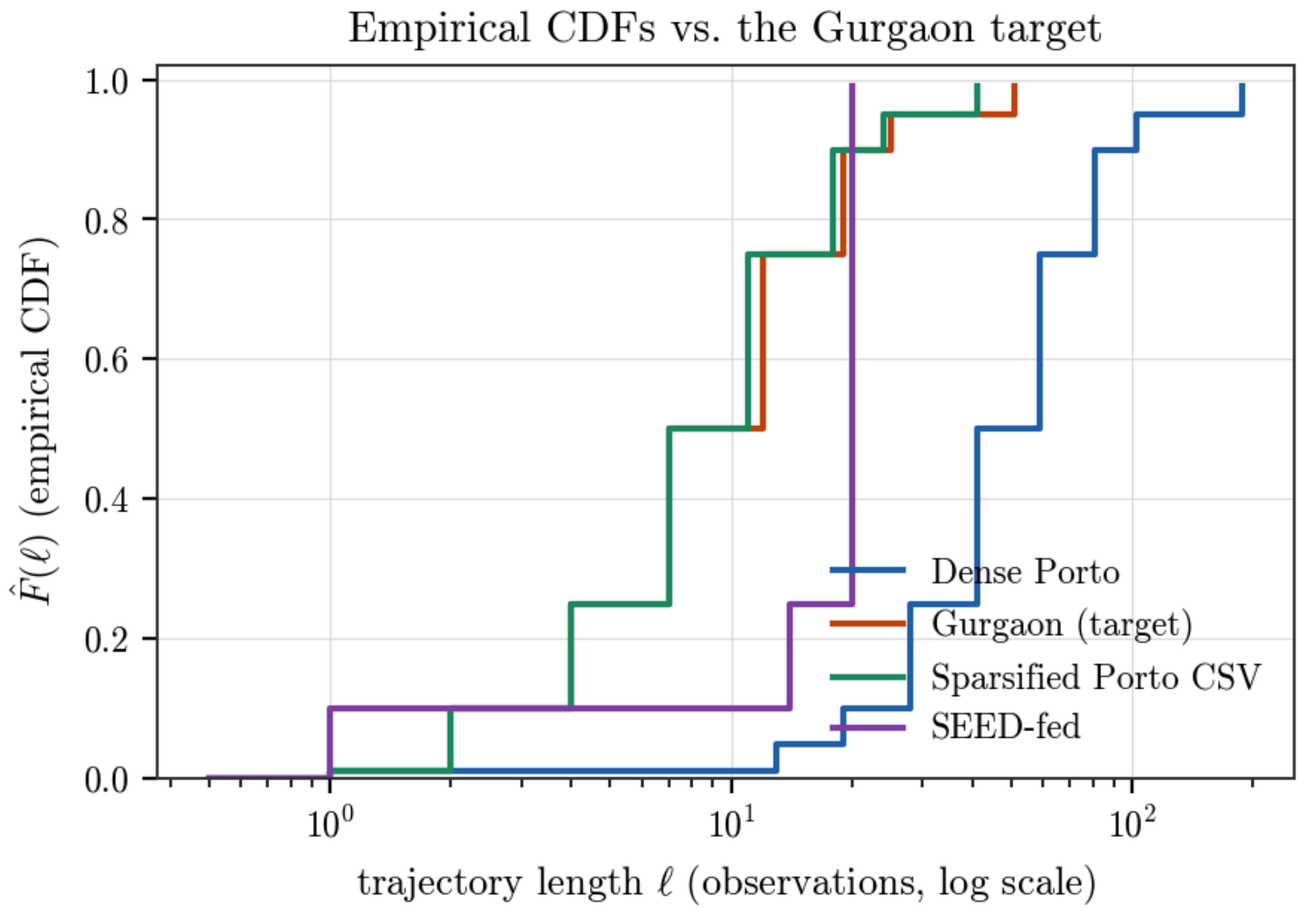
Random thinning matches aggregate rates, but Gurgaon's dates and device activity are clustered and uneven.

- Gurgaon observations occur in two blocks; sparse Porto dates are spread through the year.
- Gurgaon device activity is heavy-tailed; sparse Porto activity is much more uniform.
- Similar date coverage and entity-day density do not imply similar panel structure.



Map matching changes the sequence lengths

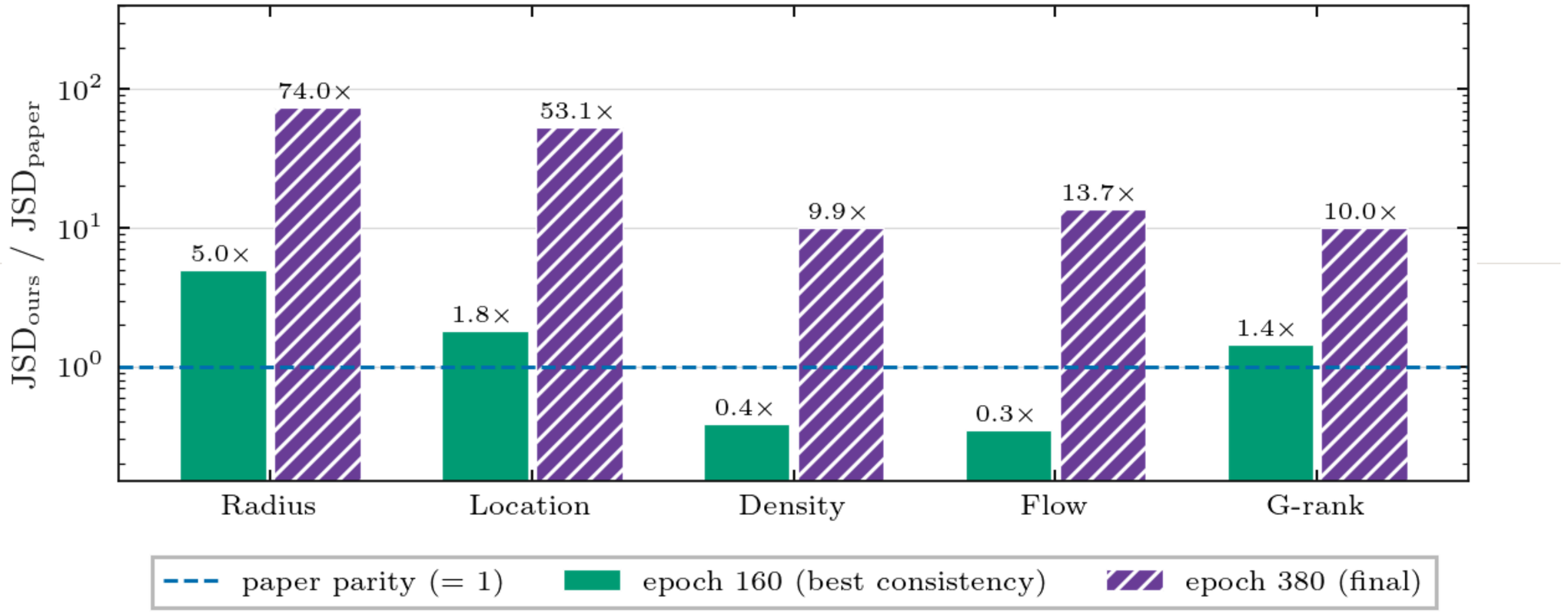
- Gurgaon and sparse CSV median lengths are 7; SEED-fed sequences have recovered median 20.
- Fixed-length resampling creates a spike at 20; exact-20 lengths are right-censored (true path length 20).
- SEED-fed KS distance to Gurgaon rises to 0.5978 after map matching



The SEED comparison uses a much smaller sample

Model and road-network settings are held constant; trip count differs substantially.

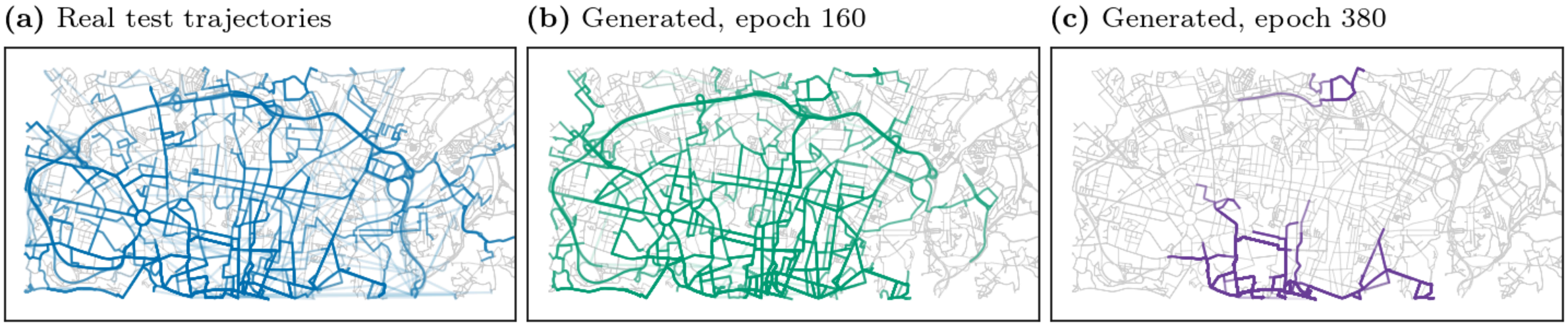
- **Dense paper dataset:** 96,896 trajectories. **Sparse run**: 8,449 (6,759 train / 1,690 test), or 8.7% of the paper sample.
- Both use 8,476 road segments, sequence length 20, 500 diffusion steps, channel size 256, learning rate 1e-4, and 50 pre-training epochs.
- Sparse run trains for 380 epochs; 1,690 trajectories are generated and evaluated every 10 epochs



Trajectory generation: connectivity rises as diversity falls

Generated paths at epoch 380 are more connected, but increasingly repeat a few corridors.

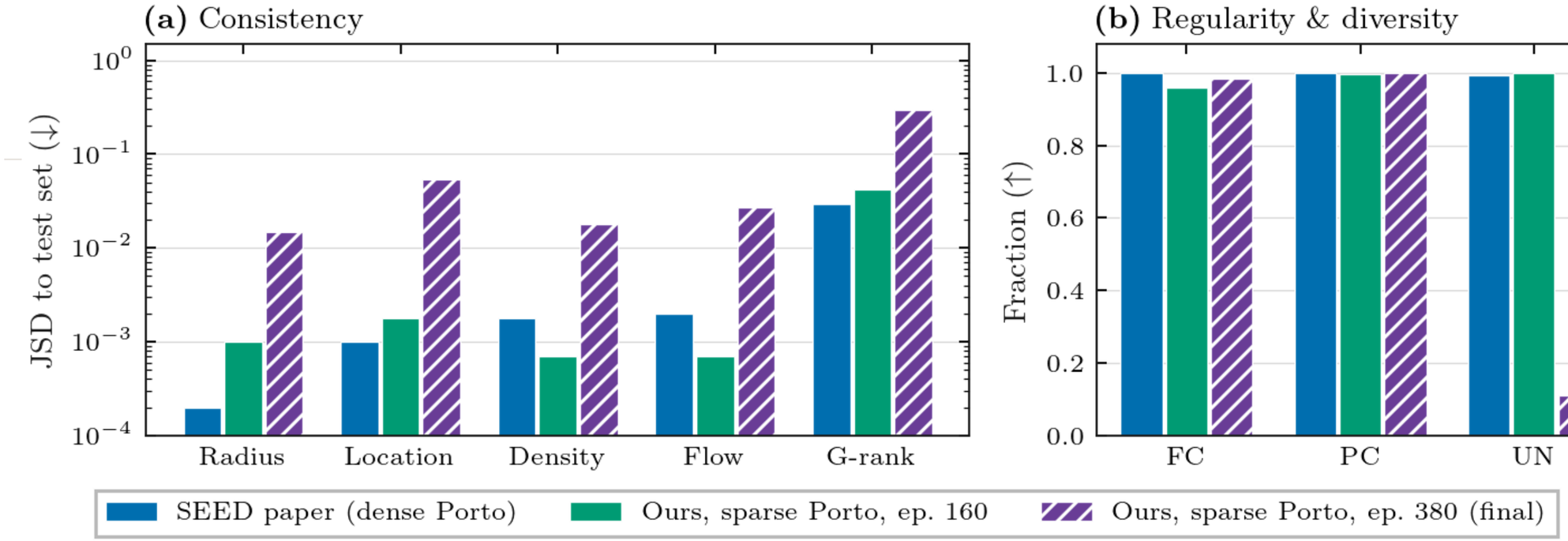
- **Real sparse-Porto test trajectories**: about 60% are road-connected after nearest-edge map matching.
- **Epoch 160:** spatially spread out; 97% connected in the displayed sample.
- **Epoch 380:** about 99% connected, but paths pile onto repeated corridors; unique-trajectory share falls to 0.111.



Epoch 160 is the useful comparison point

Consistency and diversity peak well before connectivity-based early stopping.

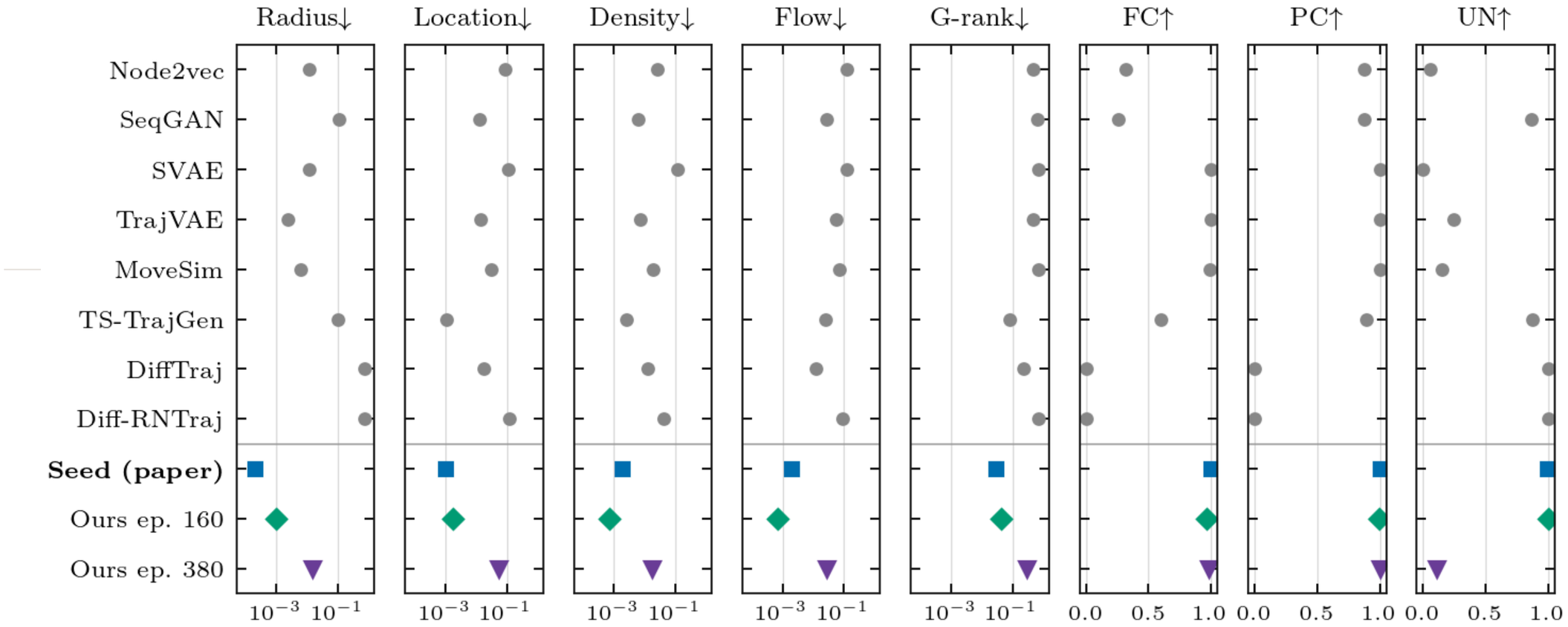
- **At epoch 160,** sparse SEED is near paper metrics and better on Density / Flow; FC 0.960, PC 0.998, UN 0.999.
- **At epoch 380,** Location / Density / Flow JSD are about an order of magnitude worse; UN is 0.111.
- **Epoch 160** was selected after inspecting test JSD, so treat it as an optimistic checkpoint, not an unbiased estimate.



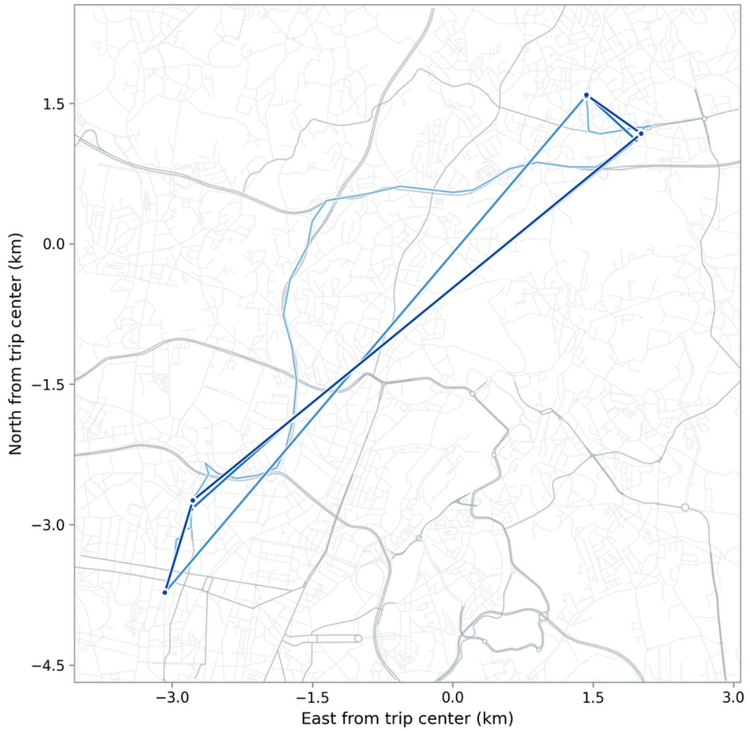
How sparse SEED ranks against Porto baselines

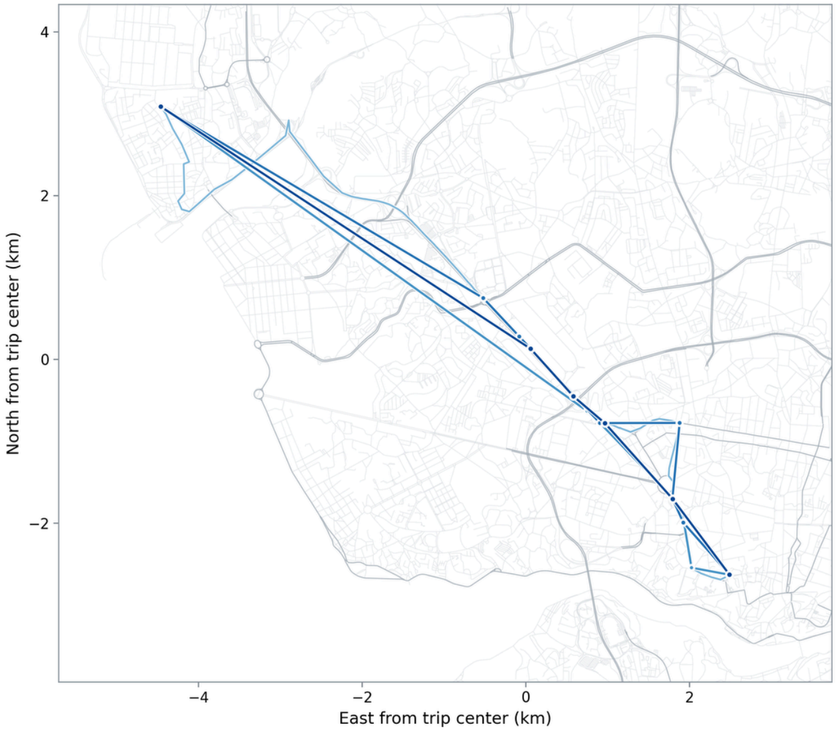
At epoch 160 it remains competitive on most consistency measures, despite the smaller training set.

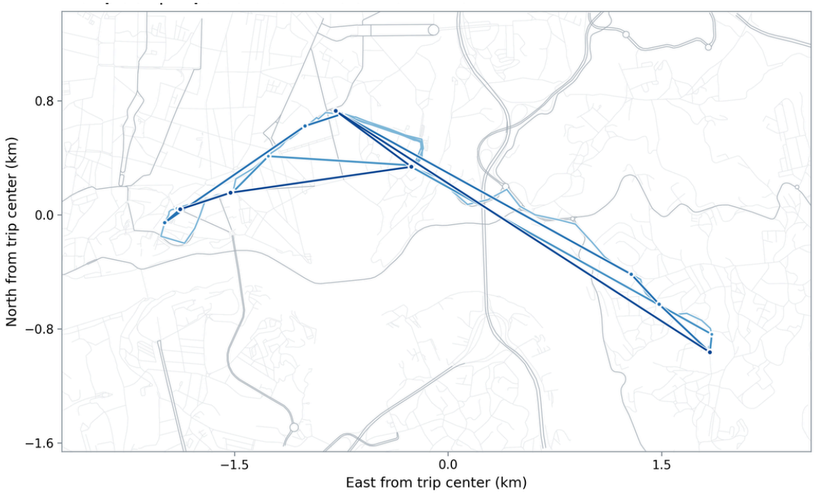
- **Epoch 160** ranks 1st of 9 for Density, Flow, and G-rank; 2nd for Location; 4th for FC and PC.
- **By epoch 380**, ranks slip to 5th Radius, 6th Location, 5th Density, and 7th diversity.
- The paper and sparse run use different test distributions, so ranks do not establish better absolute realism.

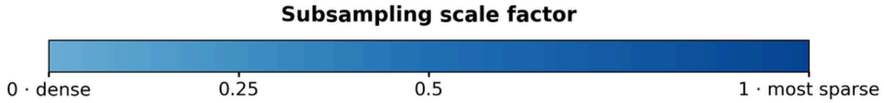


Porto Trajectory Sparsification → Gurugaon









Porto Trajectory Sparsification → Gurugaon

